

Candidate Name

Centre Number

Candidate Number



For Performance Measurement

**ZIMBABWE SCHOOL EXAMINATIONS COUNCIL**  
General Certificate of Education Advanced Level

**CHEMISTRY**  
PAPER 2

**6031/2**  
1 hour 30 minutes

**NOVEMBER 2025 SESSION**

Candidates answer on the question paper.  
Additional materials:  
Data Booklet  
Scientific Calculator

**INSTRUCTIONS TO CANDIDATES**

Write your name, centre number and candidate number in the spaces at the top of this page.

**INFORMATION FOR CANDIDATES**

The number of marks is given in brackets [ ] at the end of each question or part question

**FOR EXAMINER'S USE**

|              |  |
|--------------|--|
| <b>1</b>     |  |
| <b>2</b>     |  |
| <b>3</b>     |  |
| <b>4</b>     |  |
| <b>5</b>     |  |
| <b>6</b>     |  |
| <b>TOTAL</b> |  |

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**[Turn over**



- 1 (a) A halogen, X, reacts as shown in Fig. 1.1.

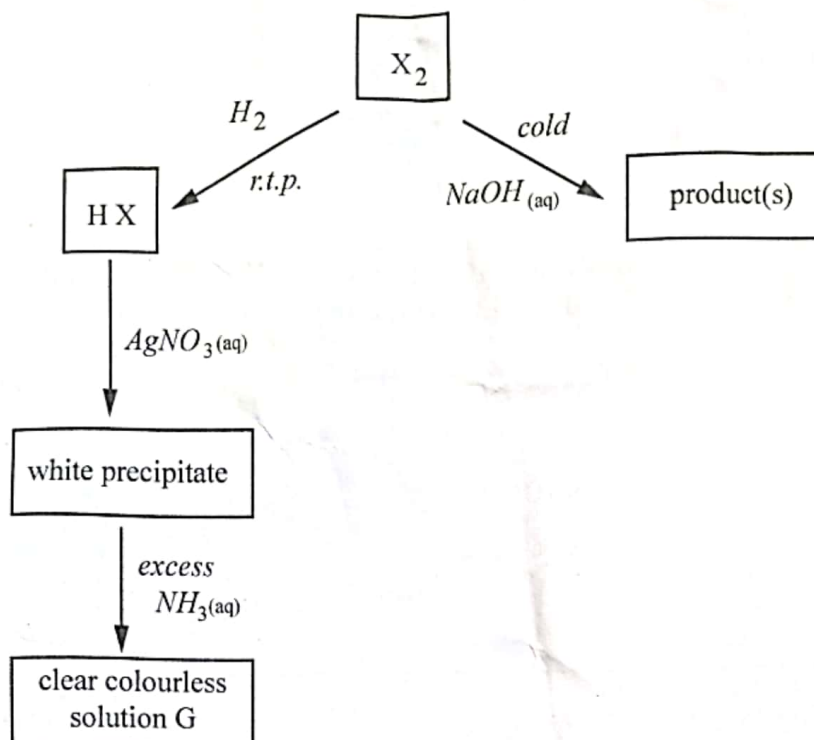


Fig. 1.1

- (i) Give the chemical formula of solution G.
- \_\_\_\_\_
- \_\_\_\_\_
- (ii) Write a balanced chemical equation to show the formation of the products when  $\text{NaOH}_{(\text{aq})}$  was added.
- \_\_\_\_\_
- (iii) Comment on the acidity of the halide, HX.
- \_\_\_\_\_
- \_\_\_\_\_
- \_\_\_\_\_

[4]



- (b) (i) Describe and explain the trend in atomic radii across Period 3 elements.

\_\_\_\_\_  
\_\_\_\_\_  
\_\_\_\_\_ [2]

- (ii) Compare the radii of  $\text{Cl}^-$  and  $\text{Cl}$ .

\_\_\_\_\_  
\_\_\_\_\_ [2]

- (c) (i) State any **one** use of  $\text{CaO}$  in Agriculture.

\_\_\_\_\_  
\_\_\_\_\_ [1]

- (ii) Write a balanced ionic equation to illustrate the action of  $\text{CaO}$  in (i).

\_\_\_\_\_  
\_\_\_\_\_ [1]

[Total: 10]

- 2 (a) Hydrazine, a rocket fuel, has the molecular formula  $\text{N}_2\text{H}_4$ .

- (i) Draw a dot and cross diagram to show bonding in hydrazine.

\_\_\_\_\_ [1]

- (ii) Describe the shape around the nitrogen atom in hydrazine.

\_\_\_\_\_ [1]

- (iii) State the bond angle between the H-N-H bonds.

\_\_\_\_\_ [1]



- (iv) Compare the boiling points of hydrazine and ammonia.

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[2]

- (b) (i) Sketch a graph of  $pV$  against  $P$  to show how helium, hydrogen ammonia and nitrogen relate to the ideal gas.

[3]

- (ii) Compare the intermolecular forces in nitrogen and ammonia.

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[2]  
[Total: 10]



- 3 Table 3.1 shows the initial rate when X and Y were reacted at different concentrations.

Table 3.1

| experiment | concentration |      | Initial<br>rate/ $\text{mol dm}^{-3} \text{s}^{-1}$ |
|------------|---------------|------|-----------------------------------------------------|
|            | X             | Y    |                                                     |
| 1          | 0.02          | 0.04 | 0.083                                               |
| 2          | 0.02          | 0.08 | 0.32                                                |
| 3          | C             | D    | 0.720                                               |
| 4          | 0.04          | 0.12 | 1.431                                               |
| 5          | 0.08          | 0.12 | 2.880                                               |

- (a) (i) Deduce the rate equation.

[3]

- (ii) Calculate, for experiment 3, the values of.

1. C,

2. D.

[2]



(b) Benzoic acid has a  $K_a$  value of  $6.3 \times 10^{-5} \text{ mol dm}^{-3}$  at  $25^\circ \text{C}$ .

(i) Write the dissociation equation for benzoic acid.

(ii) Sketch a graph to show how pH varies when  $0.02 \text{ mol dm}^{-3}$  potassium hydroxide is added to  $25 \text{ cm}^3$  of  $0.02 \text{ mol dm}^{-3}$  of benzoic acid.

[1]

(iii) Calculate the pH of  $0.02 \text{ mol dm}^{-3}$  of benzoic acid.

[1]



- (iv) Table 3.2 shows examples of acid-base indicators used during titrations.

Table 3.2.

| indicator        | pKa | Range      |
|------------------|-----|------------|
| methyl orange    | 3.7 | 3.7 – 4.4  |
| methyl red       | 5.1 | 4.8 – 6.0  |
| bromothymol blue | 7.0 | 6.0 – 7.6  |
| phenolphthalein  | 9.4 | 8.2 – 10.0 |

Identify, with a reason, the most suitable indicator for the titration of benzoic acid with potassium hydroxide.

*suitable indicator* \_\_\_\_\_

*reason* \_\_\_\_\_

[2]

[Total:10]

- 4 (a) An organic compound, T, has the structure shown.



- (i) Write the structural formulae of organic products formed when T reacts with

1.  $\text{PCl}_5$ ,

[1]

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[Turn over



2.  $I/NaOH_{(aq)}$ ,

[2]

3. Na metal.

[1]

(ii) Give any **one** observable change that occurs when **T** reacts with  $PCl_5$ .

[1]





(b) The molecular formula of an organic compound was found to be  $\text{CH}_3\text{Br}$ .

(i) Name the homologous series to which the organic compound belongs.

\_\_\_\_\_

\_\_\_\_\_

[1]

(ii) Illustrate the mechanism for the reaction between the organic compound and warm sodium hydroxide.

[3]

(iii) Suggest **one** environmental problem associated with its use.

\_\_\_\_\_

\_\_\_\_\_

[1]

[Total: 10]

5 (a) Fig. 5.1 shows the structure of an organic compound.

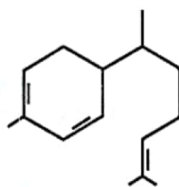


Fig 5.1

Deduce the molecular formula of the compound.

\_\_\_\_\_

[1]



- (b) (i) Identify, by means of asterisks (\*), the chiral carbons in the compound.
- (ii) State the other type of isomerism that the compound can display besides optical isomerism.
- \_\_\_\_\_ [3]
- (c) Draw the structure(s) of organic product(s) produced when the compound reacts with
- (i)  $\text{Br}_2$  in an inert solvent,
- \_\_\_\_\_ [1]
- (ii) hot concentrated  $\text{KMnO}_4(\text{aq})$ .
- \_\_\_\_\_ [3]
- (d) Describe the observable changes that occur in each of the reaction in (c).
- \_\_\_\_\_
- \_\_\_\_\_
- \_\_\_\_\_
- \_\_\_\_\_ [2]

[Total: 10]



- 6 (a) (i) State **one** alternative source of energy other than nuclear energy.

[1]

- (ii) Identify **one** possible hazard associated with nuclear accidents.

[1]

- (iii) Suggest, with a reason, the method that can be used by Urban Councils in pampers disposal.

[2]

- (b) State the biochemical importance of iron in cytochrome.

[1]



- (c) A small amount of iodine was added 25.00 cm<sup>3</sup> of tetrachloromethane and 250 cm<sup>3</sup> of water. On titration, 25.00 cm<sup>3</sup> of the aqueous layer required 6.70 cm<sup>3</sup> of 0.055 mol/dm<sup>3</sup> of sodium thiosulphate. 25.00 cm<sup>3</sup> of the organic layer required 27.20 cm<sup>3</sup> of 1.15 mol/dm<sup>3</sup> of sodium thiosulphate.

(i) State any **two** properties of a good solvent.

1. \_\_\_\_\_  
\_\_\_\_\_

2. \_\_\_\_\_  
\_\_\_\_\_

[2]

(ii) Write the partition coefficient expression.

\_\_\_\_\_

(iii) Hence calculate the partition coefficient.

[3]

[Total:10]

